Shubham Jain

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Education

- **Purdue University - West Lafayette** *Aug 2023 - Dec 2026*
  B.S in Computer Science, B.S in Artificial Intelligence, Minor in Mathematics GPA: 3.83

Technical Skills and Interests

- **Languages**: Python, Java, C/C++, SQL, R, TypeScript, HTML/CSS/JS, Assembly.
- **Web & Systems**: React.js, Node.js, Salesforce (Apex, LWCs, Data Cloud, Agentforce), Databricks, Firebase.
- **DevOps & Tools**: Git/GitHub, CI/CD (GitHub Actions), Linux/Unix, Microservices, Azure, Lakehouse.
- **Core Strengths**: Full-Stack Development, Systems Programming, Machine Learning, AI Engineering.
- **Interests**: Software Engineering, AI, Systems Engineering, Embedded Systems, FinTech, MedTech.

Experience

- **Amazon Web Services** *May 2026 - Aug 2026*
  SDE Intern Seattle, WA
  - Incoming on the Quick Suite Team.

- **Robert Half / Protiviti** *Jun 2025 - May 2026*
  Software Engineering Intern San Ramon, CA
  - Built an ML pipeline in Python and Salesforce Data Cloud, improving prediction accuracy from **72%** to **86%**.
  - Developed **20+ LWCs** and **40+ Apex classes** to enhance functionality for Protiviti consultants.
  - Configured Salesforce environments; created backend integrations, streamlining sales workflows and enabling analytics.
  - Engineered **4 autonomous AI agents** via **Agentforce** to automate record creation and editing.

- **Indiana Fever (with The Data Mine)** *Aug 2024 - May 2025*
  Machine Learning Lead: Undergraduate Data Science Researcher Indianapolis, IN
  - Led supervised learning models (XGBoost, Random Forest) predicting player performance with **80% accuracy**.
  - Designed a Databricks Lakehouse pipeline to clean, aggregate, and transform **500K+ player stats**.
  - Reduced **RMSE by 23%** through **hyperparameter tuning** with model ensembling and feature importance analysis.

- **Elastik Teams** *Jun 2022 - Aug 2022*
  Software Engineering Intern Remote
  - Built RESTful APIs with Node.js/Express.js; integrated OAuth 2.0 to **cut login latency by 20%** and improve security.
  - Redesigned microservices architecture and implemented CI/CD, reducing deployment time from **1 hr to 10 mins**.
  - Achieved 98% code coverage through comprehensive test suite with **30+ unit tests** and TDD methodologies.

Projects

- **Memory Allocator (malloc)** | C, Linux/Unix
  Custom memory management library
  - **Reimplemented malloc, free, calloc, and realloc** functions from scratch in C.
  - Designed efficient allocation algorithms to **minimize fragmentation** and improve performance.
  - Implemented memory coalescing to optimize space utilization across allocation requests.

- **Compiler** | C, Assembly Language, Lex, Yacc, Linux/Unix, Compiler Design, Systems Programming
  SimpleC compiler with lexical analysis, parsing, and code generation
  - Developed a compiler with lexical analysis, parsing, and **x86_64 assembly code** generation.
  - Implemented **semantic analysis** for type checking, scope resolution, and compile-time error detection.
  - Optimized AST-to-assembly translation for improved performance and correctness.

- **Slingshot Sumo** | Node.js, Express.js, Socket.IO, Three.js, Matter.js, HTML/CSS/JS
  Real-time multiplayer 3D physics game with lobby system and cross-platform support
  - Architected real-time multi-client web game using **Socket.IO** with lobby management, invite links, and host controls.
  - Integrated **Matter.js** physics engine for collision detection and trajectory simulation on a dynamically shrinking arena.
  - Rendered toggleable **3D/2D** views with Three.js; implemented responsive touch controls for full mobile support.